# AlcorCut Pro — How it works

How some of the app's behind-the-scenes parts work: what is kept, what is done again, and why. Written for the app's own team and testers. Status of each part is noted (**built** or **planned**).

## Sound in video exports: kept once, mixed fresh (planned)

Status: **planned, not built yet.** Goal: repeat audio-only exports in seconds, and faster video exports with sound. Sound processing is split into two layers.

**1. Kept once per clip — the slow part**

- Pulling the sound out of the video file. The sound sits in small slices between the picture data, so the phone has to read through the whole video file — gigabytes for 4K.
- The heavy processing: noise reduction, AI voice isolation, de-click, de-reverb.
- The result is kept in the video catalog, labelled with exactly what went into it: the file, the part used (the trim) and those heavy settings.

**2. Done fresh on every export — the light part (your mix)**

- EQ (more bass, more treble), volume, fades, pan, compressor.
- Music, extra tracks, ducking, audio transitions.
- These are quick sums for the processor, so they are never kept — always applied from the current settings.

**What happens when you change something**

| You change… | What happens |
|---|---|
| Bass, treble, volume, fades, the mix | Nothing needs redoing — still seconds. |
| Pitch or speed of one clip | That clip's light layer is redone from its kept sound — still fast. |
| Noise reduction / voice isolation strength on a clip | Only that clip's heavy part is redone (slow for that one clip); every other clip stays instant. |
| Trim a clip longer | Only the new part is missing — that clip is redone. |
| Add a new clip | Only the new clip is processed. |

So the sound can be tweaked freely: the editor always uses the latest settings, and only a clip changed in a heavy way costs time.

## AI noise reduction in the video editor: done once per clip (built)

- On the first export, each clip with AI noise reduction is cleaned once, at full size, and kept in the video catalog ("AI noise reduction — clip 1 of 2 (done once, reused next time)").
- That export and every later one read the cleaned clip instead of running the AI on every frame again.
- Changing the strength cleans that clip again. **Slim down** in the catalog frees the cleaned clips (they are made again when needed).
- The cleaned clip is stored at a very high bitrate so it looks the same as the AI's output — a video can't practically be stored losslessly.
- Cloud AI (Topaz) is used instead when turned on; its results are kept per clip too, so they are never paid for twice.

## AI noise reduction in the photo editor (built)

- Turning it on cleans the preview first (quick), then the full-size photo once in the background ("Cleaning the full-size photo").
- Both results are kept in the photo catalog, so reopening a photo or project never runs the AI again, and the export uses the full-size result.
- Clarity, texture, shadows, colour grading and the rest never make it run again. With RAW photos, exposure, white balance, highlights, shadows, whites and blacks do (the AI works on the developed picture) — so set those first. AI object removal and AI expand make a new picture and also need a new run.

## Long video exports in parts (built)

- Films over 4 minutes are rendered in parts of about 2½ minutes, split only at plain cuts — never inside a picture or sound transition.
- Each part is rendered on its own; if one fails it is done again on its own instead of restarting the whole film.
- The parts are then joined into one video without re-encoding (only if they don't match is the join re-encoded).

## Catalogs (built)

- **Photo catalogs** remember every photo you edit: a copy of the photo, its edit and settings, stars / flags / labels, and AI results.
- **Video catalogs** remember every clip you use: its last settings (given back when the clip is added to another project), proxies, stabilisation analysis, waveforms, thumbnails, render cache, cleaned AI clips, cloud AI results and transcripts.
- You choose how many photos / clips a catalog remembers (10 or more, or Unlimited); when it's full, the one used longest ago is let go first, with its files.
- Catalogs can be named, renamed, opened, deleted and moved (private storage, shared storage, SD card).
- Storage is lossless only: AI results as lossless PNG; RAW / TIFF copies packed losslessly when that saves space; JPEG, HEIC and video kept as they are.
- **Slim down** frees everything that can be made again and keeps photos, clips, copies and settings.

## Photo export (built)

- Exports run in a separate background process with all its memory free for the photo, so full-resolution exports with AI tools fit.
- The size you choose is always the size saved. If memory still runs short, the heaviest steps run a little smaller and the photo is brought back to your size (the app says so).
- JPEG uses the app's own encoder: full colour detail (4:4:4); at 100 % quality hardly any compression. Ultra HDR JPEGs use the phone's encoder (the gain map lives inside its format).

## Fast thumbnails in the video editor (built)

- Each clip shows a cover at once — the phone's own ready-made video thumbnail, or one quick frame.
- Filmstrips fill in coarse to fine (ends and middle first, then the gaps), several clips at a time, from the proxy when there is one.
- All thumbnails are kept in the video catalog, so a reopened project shows them straight away.

## Chromecast (built)

- The phone encodes the picture (and sound) itself and serves it on the local Wi-Fi as a live stream; Google's free Cast library connects the TV to it.
- The TV runs about 3–5 seconds behind (Chromecast streams in short segments).
- Video editor: the viewer with titles, layers and the mixed sound of all tracks. Capture: the viewfinder with the microphone.
- Switching screens keeps the TV connected; it shows the AlcorCut Pro card in between.

## Camera inside the editors (built)

- The Camera button in the photo and video editors opens the full Capture screen (every mode and pro control) over the editor, which stays open underneath.
- Photo editor: each photo goes straight into the filmstrip (without the automatic watermark). Video editor: videos and photos (as 5 s clips) go into the media list, not onto the timeline.

## Network storage — NAS (built)

- Servers are found two ways at once: those announcing themselves on the Wi-Fi (Bonjour / mDNS, as a Synology does) and anything answering on the file-sharing port (SMB, 445) in the phone's own network. An address can also be typed.
- It connects as a guest first. Only when the NAS refuses does the app ask for a user name and password, and then whether to remember them.
- A remembered sign-in is encrypted with a key kept in the phone's Android Keystore, stored in its own file that is left out of every backup, and only ever sent to that NAS. The app can use it or delete it, but never shows the password — the settings list shows server and user only.
- Picked files are copied into the app (with progress), so projects keep working when the NAS is off or the phone is away from home. A file already copied with the same size is not copied again.

## Framing on the picture (built)

- Video editor: pinch and drag on the picture set one zoom and position for the whole edit. The preview shows it live; the export applies the same zoom and shift after sizing, so the file is framed exactly as seen.
- Photo editor: the gesture is turned into the photo's crop when the fingers lift, so it's part of the edit like any crop.
- The magnifier and the playback resolution (Full to 1/16) only change the preview — the export always uses the full picture at full size.

## Team projects (built)

- A team project is three things in a shared folder (a NAS folder, or any folder the phone's folder picker reaches): **name.alcorteam** (the edit, with a save counter and who saved it), **name.alcorlock** (who has it open, refreshed every minute) and **name media/** (every file the edit uses).
- Saving copies only files the folder doesn't have yet, then writes the project. Opening copies the media down once into the app, so a second opening is quick and the edit works when the NAS is off.
- If someone else's note is less than 3 minutes old, the project opens read-only for you. If someone saved after you opened it, saving asks before writing over their version.

## Several open projects (built)

- Projects that aren't on screen are kept in memory as their full edit; switching puts the edit on screen again in a moment.
- Unsaved changes are found by comparing the edit with how it looked when it was last saved or opened; closing or exiting asks only when something really changed.

## Offline media (built)

- An offline clip is a light black stand-in video (low resolution, 10 frames a second) with the clip's own trims and settings; it remembers the file it stood for.
- Link media puts the original back when it can be read, or the file you pick — keeping the in and out points.

## Audio hardware (built)

- The chosen speaker / headphones / interface is set on every player in the editors; the chosen microphone on the audio meter, 24-bit WAV and the TV feed. Video recordings follow Android's own choice (Android doesn't let apps choose there).
- Android tells the app when a device is unplugged; if it was the chosen one, the choice is cleared, players go back to the phone's own and a notice says so.